

Two-Tower Recommender

Retrieval + ranking, trained from scratch

May 30, 2023

Headline Result

beats popularity +76% recall; ranker +7.8% nDCG

Prepared for: Sai Veda

Publishing account: Nikeshk834

Project Context

A complete, offline, deterministic recommender system: a from-scratch two-tower retrieval model, ANN candidate generation, a gradient-boosted ranker, and a leak-free temporal evaluation harness - trained on millions of implicit interactions and architected for 1B. No GPU, no network, no paid APIs.

This PDF dossier is intentionally visual and interview-ready: it captures the business problem, the technical architecture, the validation evidence, and the committed repository artifacts in a format that reads like a delivered project brief.

Repository slug

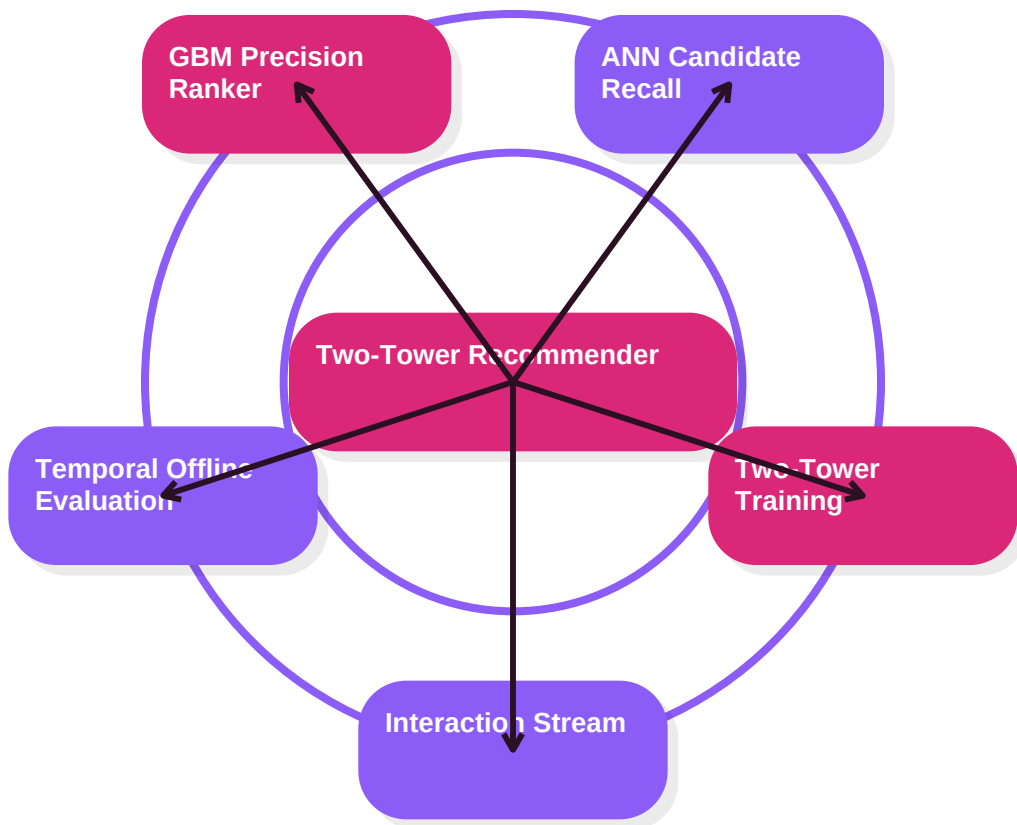
07-two-tower-recommender

Validation index

33 tests

Architecture Storyboard

Candidate recall and ranking precision are intentionally split so the serving path scales without giving up business relevance.



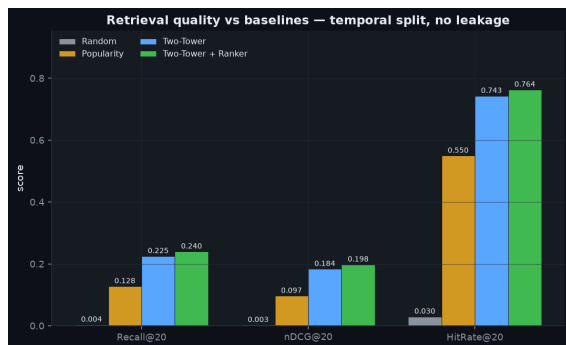
High-level focus area: Candidate recall and ranking precision are intentionally split so the serving path scales without giving up business relevance.

Validation And Visual Evidence

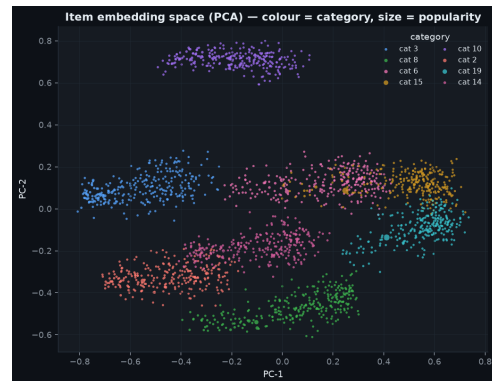
beats popularity +76% recall; ranker +7.8% nDCG

33 passing tests

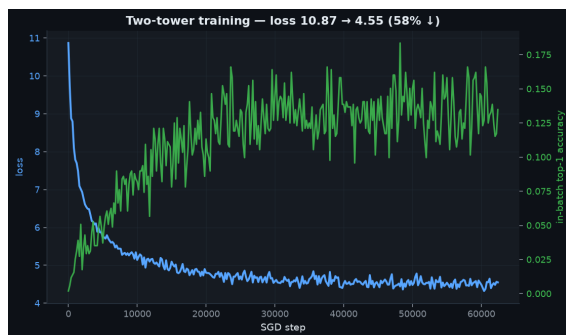
May 30, 2023



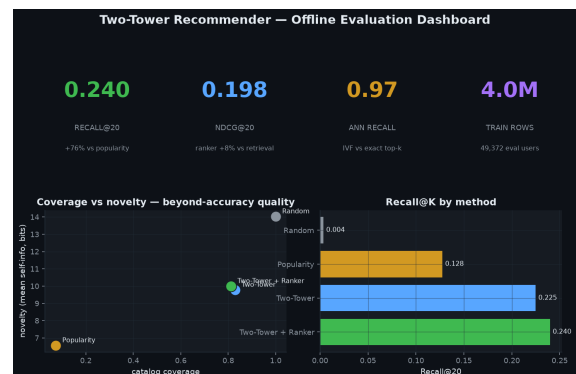
Retrieval quality vs baselines



Item embedding space (PCA), coloured by category



Two-tower training curve



Offline evaluation dashboard

Repository Handoff

Prepared for: Sal Veda

Publishing account: Nikeshk834

Project date: May 30, 2023

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Committed deliverables

- README.md - product overview, quickstart, and measured headline numbers
- ARCHITECTURE.md - detailed design and scale trade-offs
- tests/ - automated validation suite
- benchmarks/ - measured run artifacts and result tables
- assets/ - generated dashboards, charts, and screenshots
- PROJECT_DOCUMENT.pdf - styled project brief for sharing and review

This document is intentionally more visual than the repository Markdown. It is meant to read like a polished project brief: business framing first, architecture second, proof third, and delivery scope last.